



MATHEMATICS

Year 7 Final Examination

November 9, 2012

General instructions

- Working time – 90 minutes.
- Write your working (where necessary) and answers in the space provided.
- Marks may be deducted for careless or poorly arranged work.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Calculators are **NOT** to be used under any circumstances.
- All necessary working should be shown in every question.
- Attempt **all** questions.

Class (please ✓)

- ☐ 7N – Mr Lam
- ☐ 7R – Mr Ireland
- ☐ 7T – Mrs Juhn/Mr Lin
- ☐ 7H – Mrs Juhn/Mr Lowe
- ☐ 7S – Mr Berry

NAME:

Marker's use only.

SECTION	I	II	III	IV	V	Total	%
MARKS	30	16	46	13	15	120	

Section I: Number

Questions	Marks		
1. $56 + 11 + 19 + 14 =$	1	5. Evaluate $\sqrt[4]{2\,313\,441}$ if $2\,313\,441$ $= 13 \times 13 \times 13 \times 13 \times 3 \times 3 \times 3 \times 3$	2
2. Which of $<$, $=$ or $>$ would make the following statement true? $298 + 546 \dots 1\,000 - 154$	2		
3. All of the 32 rows of seats are filled for the school assembly. If there are 17 seats in each row, how many students and teachers can be seated for the assembly?	1	6. Evaluate $\frac{9}{10} - \frac{3}{4}$	2
4. (a) Draw the factor trees for 36 and 96.	2	7. What is the place value of the 6 in 4.06?	1
(b) Using the factor tree, or otherwise, find preferably in index form: i. highest common factor (HCF) of 36 and 96. ii. lowest common multiple of 36 and 96.	1 1	8. Express 0.124 as the simplest fraction. 9. Express $\frac{21}{15}$ as a decimal. 10. Calculate the value of $2.903\,2 + 83.184$ 11. Calculate the value of 5.205×100	2 2 1

- | | | |
|---|--|---------------------------------|
| <p>12. The sum of two fractions is $\frac{13}{12}$. If one fraction is $\frac{1}{4}$, what is the other one?</p> <p>.....</p> <p>.....</p> | <p>15. Evaluate $(-6)^2$.</p> <p>.....</p> | <p>2</p> <p>1</p> |
| <p>13. What is the sum of $\frac{1}{5}$ and half of $\frac{1}{5}$?</p> <p>.....</p> <p>.....</p> | <p>16. Convert $12\frac{3}{4}\%$ to a decimal.</p> <p>.....</p> <p>.....</p> | <p>2</p> <p>2</p> |
| <p>14. Evaluate $-4 \times (5 - 8)$.</p> <p>.....</p> <p>.....</p> | <p>17. There are 1 200 students in a school and 150 of them are in Year 7. What percentage of the student population are in Year 7?</p> <p>.....</p> <p>.....</p> | <p>1</p> <p>2</p> |

Section II: Space & Geometry

Questions

Marks

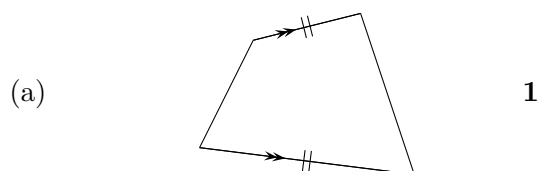
1. Fill in the blank spaces with the correct word from the list of words provided.

- Each space will only contain one word.
- Not all words are necessarily used; words may be used more than once.
- ☠ Incorrectly spelt words will result in no marks being awarded. ☠

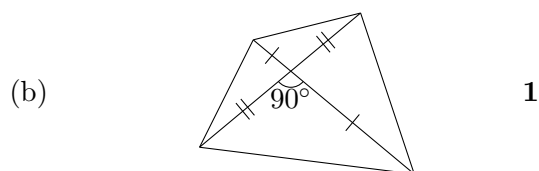
90° 180° 360° alternate bisect bysect cointerior equal
 equilateral equilateral isosceles isosceles kite parallelogram
 parallelogram parallelogram perpendicular rhombus rombus
 scalene scaline square trapezium

- (a) Adjacent complementary angles sum to 1
- (b) The minimal property that qualifies a quadrilateral to become a is to have one pair of opposite sides parallel. 1
- (c) To construct a 30° angle using compass and ruler, first construct an triangle, and then one of its angles. 2

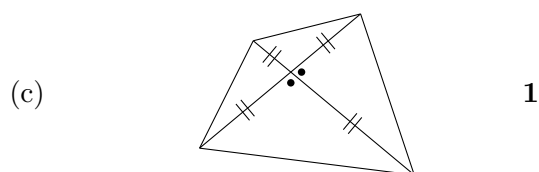
2. Name the special quadrilateral that has these minimal properties.



Shape:

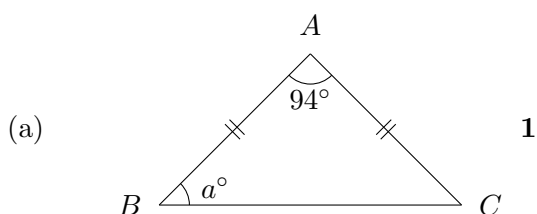


Shape:

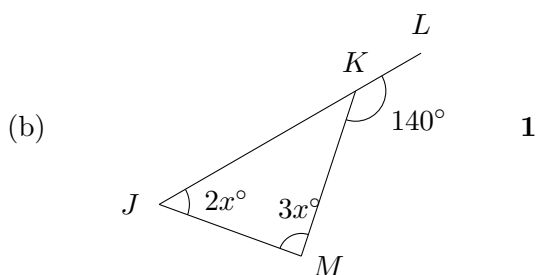


Shape:

3. Find the value of the pronumerals in the following diagrams.

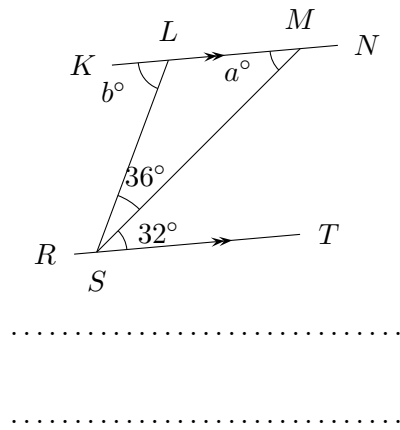


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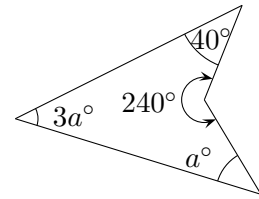
.....

(c)



2

(d)

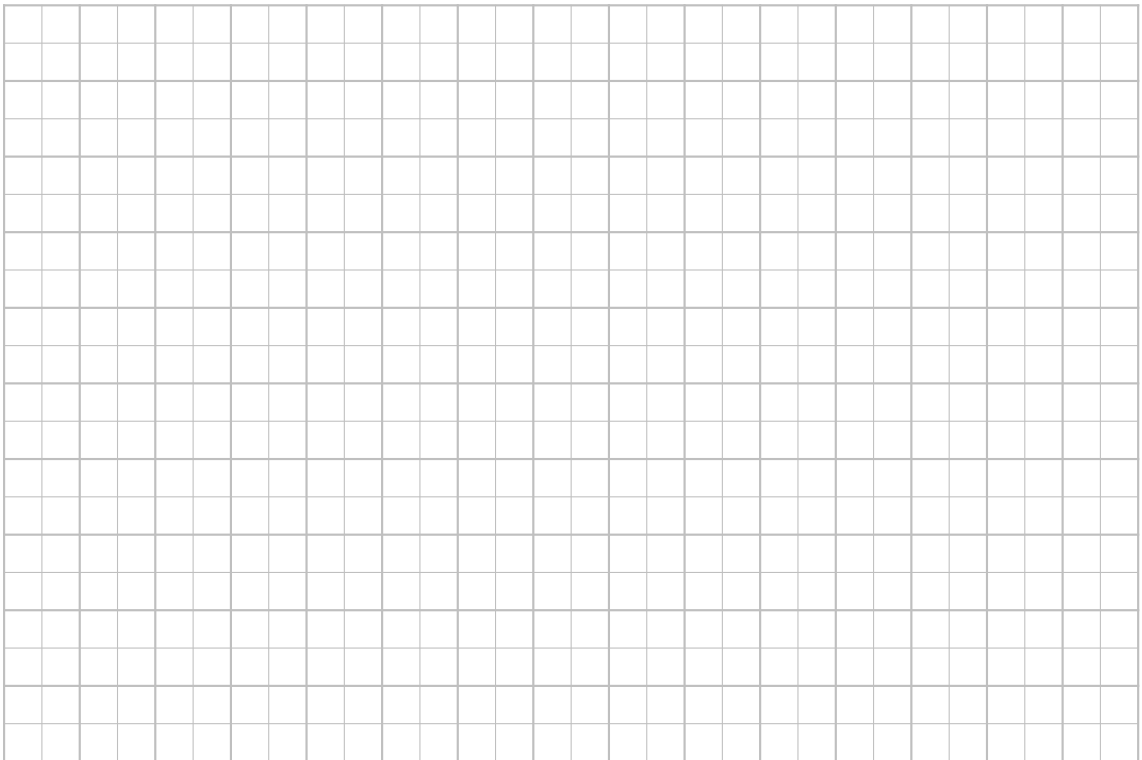


2

4. Using ruler and a pair of compasses *only*, construct a 15° angle.

3

Hint Commence by constructing a 60° angle.



Section III: Patterns & Algebra

Questions	Marks	
1. Write an algebraic expression for these sentences.		4. Simplify fully:
(a) The difference between x and y , where y is greater than x .	1	(a) $x \times x \times x \times x \times y \times y \times y$ 1
.....		
(b) Three times the number of children in the library if k leave the library out of a total of s students.	1	(b) $4h^7 \div 8h^3$ 2
.....		
2. Simplify fully:		5. If $p = 4$, $q = -6$, $r = 3$ and $t = -1$, find the value of 2
(a) $-2y \times 8z \times \frac{1}{2}w$	2	$q^2 + 6p + rt$
.....		
(b) $13 + 2ab - 6ab - 1$.	2	
.....		6. Solve the following equations, showing all working and expressing the answer in simplest form.
(c) $3d^2 - 8d^2 + 2d^2$.	1	(a) $2x + 3 = 1$. 1
.....		
3. Expand and fully simplify:		
(a) $-3(x + 2)$	1	
.....		
(b) $2c(5b - 3a)$	1	(b) $3(z + 7) = 18$. 2
.....		
(c) $3(d + 2) - (d + 8)$	2	
.....		
.....		(c) $6k - 8 = 9k + 10$. 2
(d) $5(3 - 2x) + 5x - 3$	2	
.....		
.....		

(d) $\frac{6x+5}{7} - 3 = 4.$

3

.....

.....

.....

.....

(e) $\frac{9g}{2} + 8 = -5.$

2

.....

.....

.....

7. Fully factorise the following expressions:

(a) $3x + 12xy$

1

.....

(b) $-6a^2 + 18a^3$

2

.....

(c) $5u^3 - 20u^2 + 10u^4v^2.$

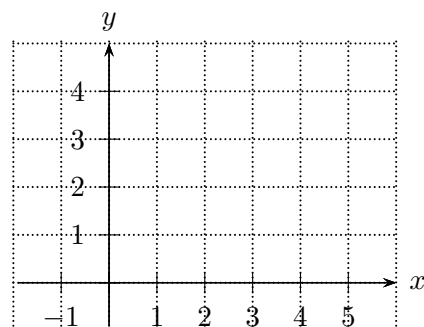
2

.....

8. (a) Complete the table of values for $y = -\frac{1}{2}x + 2.$

x	-1	0	1	2
y				

(b) Using the table of values, draw the line $y = -\frac{1}{2}x + 2.$

2

9. Write down the rule for y in terms of $x.$

(a)

x	-2	-1	0	1	2
y	7	5	3	1	-1

2

.....

(b)

x	-4	-2	0	2	4
y	2	5	8	11	14

2

.....

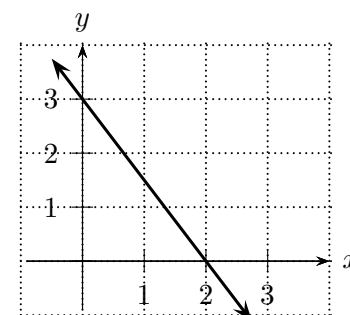
(c)

x	-2	-1	0	1	2
y	5	2	1	2	5

2

.....

10. By drawing a table of values, find the equation of the straight line given the following graph?

3

x			
y			

.....

Section IV: Measurement

Questions

Marks

1. Complete the following conversions:

(a) $3 \text{ kg} = \dots\dots\dots \text{ mg}$ **1**

(b) $2\,300 \text{ kL} = \dots\dots\dots \text{ L}$ **1**

(c) $197 \text{ mm} = \dots\dots\dots \text{ m}$ **1**

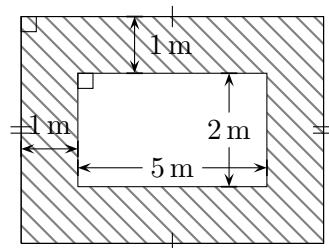
2. Calculate the time that is 3 hours, 23 minutes and 59 seconds after 1:26:03 am as 12 hour time. **2**

3. Anvesh runs at 9 metres per second for T seconds, and covers 468 m. For how long did he run? **2**

4. Find the cost of travelling in a taxi for 36 km if it costs \$2.33 per kilometre. **1**

5. Find the area of the shaded region below.

(a)



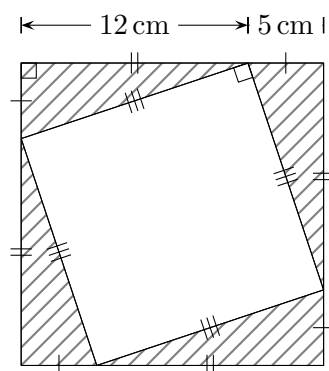
3

.....

.....

.....

(b)



2

.....

.....

.....

Section V: Working Mathematically

Questions	Marks	
1. Fully factorise $2x(x - 1) + 3(x - 1)$.	1	4. Find the LCM of $2^2 \times 3 \times 5^4$ and $2^3 \times 3^2 \times 7^2$ in index notation.
.....		
2. Which is the more economical option for buying chocolate? 250 g for \$4.50 or 1 050 g for \$12?	2	
.....		5. If 200 m^2 of grass is required for grazing three (3) healthy sheep, then how many healthy sheep can 2.5 hectares support?
.....		
3. If it takes 4 employees 5 days to clean up the rubbish alongside a 180 km stretch of freeway, how many people will be needed to clean 252 km of freeway in 4 days? (Assume they all work at the same pace.)	3	
.....		
.....		
.....		
.....		
.....		

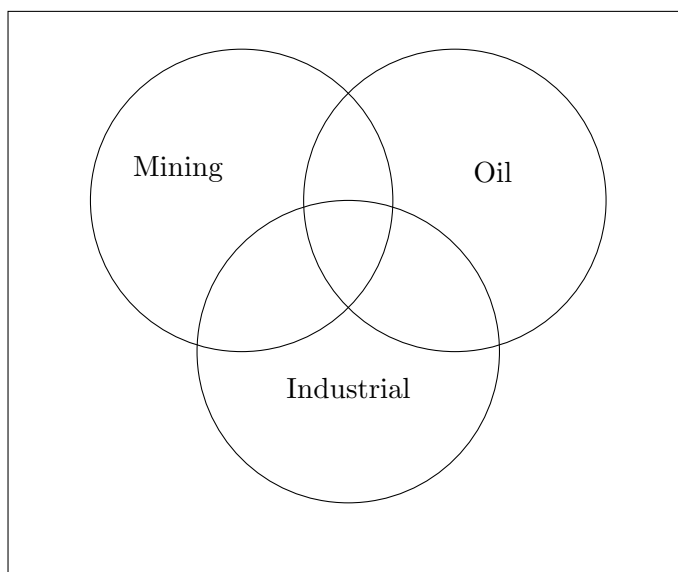
5. One hundred and thirty three people are attending an investment seminar. All 133 people are already invest in either mining, oil or industrial shares or a combination of these, with 96 members own mining shares, 70 have oil shares and 66 have industrial shares.

Of those who own mining shares, 40 also own oil shares and 45 also have industrial shares.

The number of people who own oil shares and industrial shares is 28.

- (a) Using the information given, fill in all sections of the Venn diagram provided. 3

Hint: Let the number of people who own all three types of shares be x , and fill in blank sections of the Venn diagram in terms of x .



- (b) Hence or otherwise, find the number of people who own all three types of shares. 2

.....

.....

End of paper.

Suggested Solutions

Section I: Number

1. (1 mark)

$$56 + 11 + 19 + 14 = 100$$

2. (2 marks)

- ✓ [1] for working
- ✓ [1] for correct symbol, based on working.

$$298 + 546 < 1\,000 - 154$$

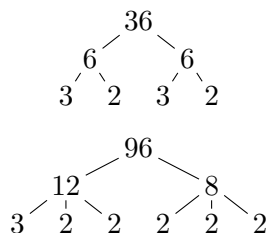
$$\begin{array}{r} 11 \\ 298 \\ + 546 \\ \hline 844 \end{array} \qquad \begin{array}{r} 1000 \\ - 154 \\ \hline 846 \end{array}$$

3. (1 mark)

$$\begin{array}{r} 17 \\ \times 32 \\ \hline 34 \\ 51 \\ \hline 544 \end{array}$$

4. (a) (2 marks)

- ✓ [1] for each correct factor tree.



$$36 = 2^2 \times 3^2$$

$$96 = 2^5 \times 3$$

(b) i. (1 mark)

$$\text{hcf}(36, 96) = 3 \times 2^2 = 12$$

ii. (1 mark)

$$\text{lcm}(36, 96) = 3^2 \times 2^5 = 288$$

5. (2 marks)

- ✓ [1] for dividing indices by 4.
- ✓ [1] for simplified answer.

$$\sqrt[4]{2\,313\,441} = \sqrt[4]{13^4 \times 3^4} \\ = 13 \times 3 = 39$$

6. (2 marks)

- ✓ [-1] for each error.
- ✓ [0] marks for entire part if common denominator not used.

$$\frac{9}{10} \times \frac{2}{2} - \frac{3}{4} \times \frac{5}{5} = \frac{18}{20} - \frac{15}{20} = \frac{3}{20}$$

7. (1 mark) hundredths

8. (2 marks)

- ✓ [-1] for each error.

$$0.124 = \frac{124}{1\,000} \stackrel{\div 4}{=} \frac{31}{250}$$

9. (2 marks)

- ✓ [1] for $\frac{21}{15} = 1 + \frac{2}{5}$.
- ✓ [1] for correct final answer.

$$\frac{21}{15} \stackrel{\div 3}{=} \frac{7}{5} = 1 + \frac{2}{5} = 1.4$$

10. (2 marks)

- ✓ [-1] for each error.

$$\begin{array}{r} 1 \\ 2.9032 \\ + 83.184 \\ \hline 86.0872 \end{array}$$

11. (1 mark)

$$5.205 \times 100 = 520.5$$

12. (2 marks)

- ✓ [1] for $\frac{1}{4} + x = \frac{13}{12}$
 ✓ [1] for correct answer.

Let x be the other fraction.

$$\begin{aligned} \frac{1}{4} + x &= \frac{13}{12} \\ -\frac{1}{4} & \quad -\frac{1}{4} \\ x &= \frac{13}{12} - \frac{1}{4} \times \frac{3}{3} \\ &= \frac{10}{12} = \frac{5}{6} \end{aligned}$$

13. (2 marks)

- ✓ [-1] for each error.

$$\begin{aligned} \frac{1}{5} + \frac{1}{2} \times \frac{1}{5} &= \frac{1}{5} \left(1 + \frac{1}{2} \right) \\ &= \frac{1}{5} \times \frac{3}{2} = \frac{3}{10} \end{aligned}$$

14. (1 mark)

$$-4(5 - 8) = -4(-3) = 12$$

15. (1 mark)

$$(-6)^2 = 36$$

16. (2 marks)

- ✓ [1] for correct location of decimal dot.
 ✓ [1] for all correct digits: 1275

$$12\frac{3}{4}\% = 0.1275$$

17. (2 marks)

- ✓ [1] for dividing 150 by 1 200
 ✓ [1] for final answer.

$$\frac{150}{1\,200} \div \frac{15}{15} = \frac{1}{8} = 12.5\%$$

Section II: Space & Geometry

1. (1 mark) each

- (a) 90°
 (b) trapezium
 (c) equilateral, bisect

2. (1 mark) each. Reasons not required (but given for future reference)

- (a) **parallelogram**
 (one pair of opposites sides equal & parallel)
 (b) **rhombus**
 (diagonals bisect each other and are perpendicular)
 (c) **square**
 (diagonals bisect each other, at right angles and are equal)

3. Reasons not required but are given for future reference.

(a) (1 mark)

$$\begin{aligned} 2a + \frac{94^\circ}{-94^\circ} &= \frac{180^\circ}{-94^\circ} \\ 2a &= 86^\circ \\ a &= 43^\circ \end{aligned}$$

($\angle$ sum of $\triangle$)

(b) (1 mark)

$$\begin{aligned} 5x &= 140^\circ \\ x &= \frac{140}{5} = 28^\circ \end{aligned}$$

(exterior $\angle$ of $\triangle$)

(c) (2 marks)

$$\begin{aligned} a &= 32^\circ \\ b &= 32^\circ + 36^\circ = 68^\circ \end{aligned}$$

(alternate $\angle$, $KN \parallel RT$)

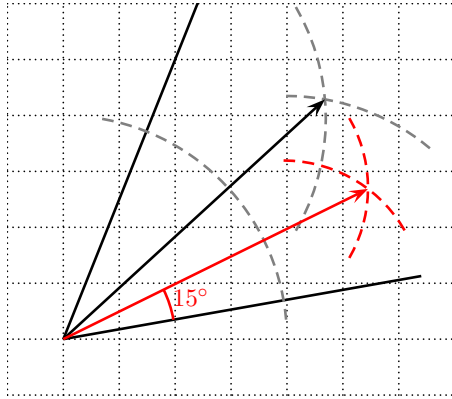
(d) (2 marks)

$$\begin{aligned} 3a + a + 240^\circ + 40^\circ &= 360^\circ \\ 4a + \frac{280}{-280} &= \frac{360}{-280} \\ 4a &= 80 \\ \therefore a &= 20^\circ \end{aligned}$$

($\angle$ sum of quadrilateral)

4. (3 marks)

- ✓ [1] for constructing $\angle 60^\circ$ (equilateral $\triangle$)
- ✓ [1] for bisecting $\angle 60^\circ$ to $\angle 30^\circ$
- ✓ [1] for bisecting $\angle 30^\circ$ again to obtain 15°



Section III: Patterns & Algebra

1. (a) (1 mark)

$$y - x$$

(b) (1 mark)

$$3(s - k)$$

2. (a) (2 marks)

- ✓ [1] for numbers (incl. negative sign)
- ✓ [1] for correct pronumerals.

$$-2y \times 8z \times \frac{1}{2}w = -8wyz$$

(b) (2 marks)

- ✓ [1] for numbers (incl. negative sign)
- ✓ [1] for correct pronumerals.

$$13 + 2ab - 6ab - 1 = 12 - 4ab$$

(c) (1 mark)

$$3d^2 - 8d^2 + 2d^2 = -3d^2$$

3. (a) (1 mark)

$$-3(x + 2) = -3x - 6$$

(b) (1 mark)

$$2c(5b - 3a) = 10bc - 6ac$$

(c) (2 marks)

- ✓ [1] for correct expansion.
- ✓ [1] for correct simplification.

$$3(d + 2) - (d + 8) = 3d + 6 - d - 8 = 2d - 2$$

(d) (2 marks)

- ✓ [1] for correct expansion.
- ✓ [1] for correct simplification.

$$5(3 - 2x) + 5x - 3 = 15 - 10x + 5x - 3 = 12 - 5x$$

4. (a) (1 mark)

$$x \times x \times x \times x \times y \times y \times y = x^4y^3$$

(b) (2 marks)

- ✓ [1] for number.
- ✓ [1] for index.

$$4h^7 \div 8h^3 = \frac{1}{2}h^4$$

5. (2 marks)

- ✓ [1] for correct substitution.
- ✓ [1] for final answer.

$$q^2 + 6p + rt = (-6)^2 + 6(4) + 3(-1) = 36 + 24 - 3 = 57$$

6. (a) (1 mark)

$$2x + \frac{3}{-3} = \frac{1}{-3}$$

$$2x = -2$$

$$x = -1$$

(b) (2 marks)

- ✓ [-1] for each error.

$$\frac{3}{\div 3}(z + 7) = \frac{18}{\div 3}$$

$$z + \frac{7}{-7} = \frac{6}{-7}$$

$$z = -1$$

(c) (2 marks)

✓ [-1] for each error.

$$\frac{6k}{-6k} - 8 = \frac{9k}{-6k} + 10$$

$$\frac{-8}{-10} = \frac{3k}{-10} + \frac{10}{-10}$$

$$3k = -18$$

$$k = -6$$

(d) (3 marks)

✓ [2] for obtaining $x = \frac{44}{6}$, with fully correct working.✓ [1] for simplifying fraction to $\frac{22}{3}$.

$$\frac{6x+5}{7} - \frac{3}{+3} = \frac{4}{+3}$$

$$\frac{6x+5}{7} = \frac{7}{\times 7}$$

$$6x + 5 = \frac{49}{-5}$$

$$6x = 44$$

$$x = \frac{44}{6} = \frac{22}{3}$$

(e) (2 marks)

✓ [-1] for each error.

$$\frac{9g}{2} + 8 = \frac{-5}{-8}$$

$$\frac{9g}{2} = \frac{-13}{\times 2}$$

$$9g = -26$$

$$g = -\frac{26}{9}$$

7. (a) (1 mark)

$$3x + 12xy = 3x(1 + 4y)$$

(b) (2 marks)

✓ [1] for each correct factor.

✓ [-1] if not fully factorised.

$$-6a^2 + 18a^3 = -6a^2(1 - 3a)$$

(c) (2 marks)

✓ [1] for each correct factor.

✓ [-1] if not fully factorised.

$$5u^3 - 20u^2 + 10u^4v^2 = 5u^2(u - 4 + 2u^2v^2)$$

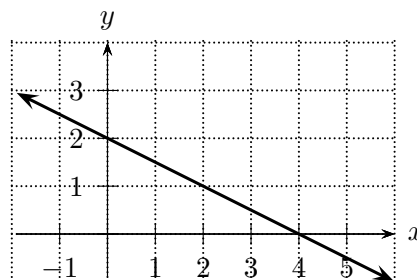
8. (a) (2 marks)

✓ [-1] for every two errors.

x	-1	0	1	2
y	2.5	2	1.5	1

(b) (2 marks)

✓ [-1] for every two errors in plotting.



9. (a) (2 marks)

✓ [1] for correct gradient (m)✓ [1] for correct y intercept (b)

$$y = -2x + 3$$

(b) (2 marks)

✓ [1] for correct gradient (m)✓ [1] for correct y intercept (b)

$$y = 1.5x + 8$$

(c) (2 marks)

✓ [1] quadratic term (x^2)

✓ [1] for correct shift.

$$y = x^2 + 1$$

10. (3 marks)

✓ [1] for correct table of values.

✓ [1] for correct gradient (m)✓ [1] for correct y intercept (b)

x	0	1	2
y	3	1.5	0

$$y = -1.5x + 3$$

Section IV: Measurement

1. (a) (1 mark)

$$3 \text{ kg} = 3\,000\,000 \text{ mg}$$

- (b) (1 mark)

$$2\,300 \text{ kL} = 2\,300\,000 \text{ L}$$

- (c) (1 mark)

$$197 \text{ mm} = 0.197 \text{ m}$$

2. (2 marks)

$$\begin{array}{r} 3 : 23 : 59 \\ + 1 : 26 : 03 \\ \hline 4 : 50 : 02 \\ \therefore 4 : 50 : 02 \text{ am} \end{array}$$

3. (2 marks)

- ✓ [1] for correctly recalling speed/dist/time formula.
- ✓ [1] for final answer, with correct units.

$$\begin{aligned} \text{speed} &= \frac{\text{dist}}{\text{time}} \\ 9 &= \frac{468}{T} \\ 9T &= 468 \\ T &= \frac{468}{9} = 52 \text{ seconds} \end{aligned}$$

4. (1 mark)

$$\begin{array}{r} 3\,6 \\ \times 2.3\,3 \\ \hline 1\,0\,8 \\ 1\,0\,8 \\ 7\,2 \\ \hline 8\,3.8\,8 \end{array}$$

5. (a) (3 marks)

- ✓ [1] for area of inner rectangle.
- ✓ [1] for area of outer rectangle.
- ✓ [1] for area of shaded area.

$$A = 28 - 10 = 18 \text{ m}^2$$

- (b) (2 marks)

- ✓ [1] for area of one triangle.
- ✓ [1] for area of shaded area.

$$\begin{aligned} A &= 4 \times \left(\frac{1}{2} \times 12 \times 5 \right) \\ &= 2 \times 12 \times 5 = 120 \text{ cm}^2 \end{aligned}$$

Section V: Working Mathematically

1. (1 mark)

$$2x(x-1) + 3(x-1) = (x-1)(2x+3)$$

2. (2 marks)

- ✓ [1] for showing working.
- ✓ [1] for correct choice.

Grams per dollar for smaller size: $\frac{250 \text{ g}}{\\$4.5} = \frac{2\,500 \text{ g}}{\\$45}$	Grams per dollar for larger size: $\frac{1\,050 \text{ g}}{\\$12}$
$\begin{array}{r} 55 \\ 45 \overline{) 2500} \\ \underline{2250} \\ 250 \\ \underline{225} \\ 25 \end{array}$	$\begin{array}{r} 87 \\ 12 \overline{) 1050} \\ \underline{960} \\ 90 \\ \underline{84} \\ 6 \end{array}$
i.e. 55.55 g per \$1	i.e. 87 g per \$1

Hence the larger size (1 050 g for \$12) is cheaper.

3. ✓ [2] for logical working.
 ✓ [1] for final answer.

- 4 employees/5 days for 180 km
- 4 employees/1 day for 36 km ($\div 5$)
- 1 employee/1 day for 9 km
- 1 employee/28 days for 252 km

$$\begin{array}{r} 28 \\ 9 \overline{) 252} \\ \underline{180} \\ 72 \\ \underline{72} \\ 0 \end{array}$$

- 7 employees/4 days for 252 km

Hence 7 employees required.

4. (1 mark)

$$2^3 \times 3^2 \times 5^4 \times 7^2$$

5. (3 marks)

- ✓ [1] for correct conversion of hectares to m^2 .
- ✓ [1] for setting up ratio.
- ✓ [1] for final answer.

$$\frac{3 \text{ sheep}}{20 \text{ m}^2 \times 25\,000} = \frac{x \text{ sheep}}{25\,000 \text{ m}^2}$$

$$x = \frac{3 \times 25\,000}{20} = 375$$

6. (a) (3 marks)

- ✓ [1] placing x (or 14) in $M \cap O \cap I$.
- ✓ [1] for placing $40-x$, $45-x$, $28-x$ (or equivalent correct answers) in middle regions.
- ✓ [1] for placing $11+x$, $x+2$, $x-7$ in outer regions.
- Given x is the number of people owing all three, those owning
 - Mining (M) + Oil (O) (but not all three) will be $40-x$
 - Mining + Industrial (I) (but not all three) will be $45-x$.
 - O and I (but not all three) will be $28-x$

- Now, $M = 96$ in total.

$$\overbrace{(40-x)}^{M \& I \text{ only}} + x + \overbrace{(45-x)}^{M \& O \text{ only}} = 85 - x$$

$$\therefore \text{Mining only} = 96 - (85 - x) = 11 + x$$

- $O = 70$ in total.

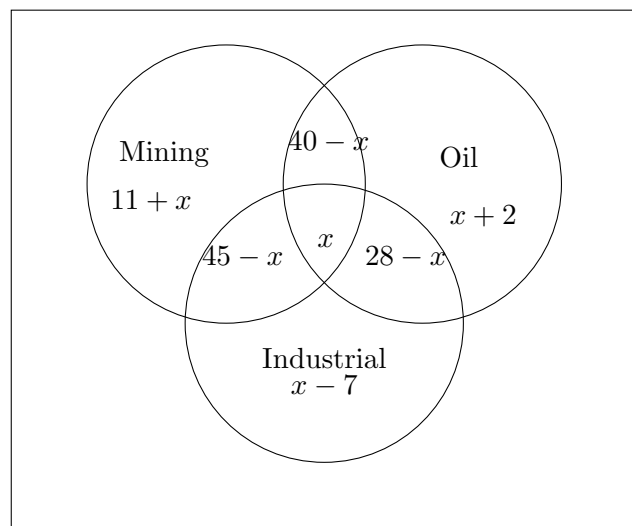
$$(40-x) + x + \overbrace{(28-x)}^{O \& I \text{ only}} = 68 - x$$

$$\therefore \text{Oil only} = 70 - (68 - x) = x + 2$$

- $I = 66$ in total.

$$(45-x) + x + (28-x) = 73 - x$$

$$\therefore \text{Indust only} = 66 - (73 - x) = x - 7$$



- (b) (2 marks)

- ✓ [1] for forming equation.
- ✓ [1] for solving equation to obtain $x = 4$.

Summing the information in the Venn diagram,

$$\begin{array}{rrr} & x & +2 \\ + & -x & +28 \\ + & x & -7 \\ + & x & \\ + & -x & +40 \\ + & -x & +45 \\ + & x & +11 \\ \hline & x & +129 \end{array}$$

$$\therefore x + 119 = 133$$

$$x = 14$$